

PLINK GWAS Population Substructure Answers

Table 1

	Un-adjusted	PC 1	PC 1-2	PC 1-3	PC 1-4	PC 1-5	PC 1-6	PC 1-7	PC 1-8	PC 1-9	PC 1-10
λ	1.2.1	1.085	1.026	1.033	1.040	1.050	1.043	1.021	1.036	1.043	1.051

Answer 1: Yes, see tables below. It is best to include to two PC components in the analysis, however the lambda is still inflated. Since we are analyzing three unique populations inclusion of PCs did not adequately control for substructure. If you compare the QQ plots below you can see that for this dataset the most significant SNPs were changed minimally when we adjusted for substructure but some of the moderately significant SNPs became less significant after adjustment. However, in some situations the p-values can become smaller.

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
8	rs4571722	60326734	T	ADD	242	0.04126	-7.436	1.04E-13
4	rs10008252	179853616	G	ADD	244	0.1665	-6.639	316E-11

CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
8	rs4571722	60326734	T	ADD	242	0.04382	-7.237	4.59E-13
4	rs10008252	179853616	G	ADD	244	0.13070	-6.707	1.99E-11